

SWP391- Project AI Usage Report				
Subject Code	SWP391			
Subject Name	Software Development Project			
Class Code	SE2018			
Semester	SU26			
Lecturer Name	LinhNDM			
Group Code	Group 5 (swp391_nhom5)			
Project Title	TutorFinder EdTech Platform - Online Tutor Booking & Learning Marketplace			
List of Student				
No	StudentCode	StudentName	Role In Group	AI Tool Usaged
1	SE181001	Nguyễn Văn Phúc	Domain Owner: Auth & Profiles (17 UCs, Entities & Controllers)	ChatGPT (GPT-4o), Antigravity,Gemini Pro
2	HE186547	Nguyễn Hoàng Hưng	Team Leader / Domain Owner: Search & Booking (18 UCs, UML Specialist)	ChatGPT (GPT-4o), Antigravity,Gemini Pro
3	HE180202	Kiều Duy Doanh	Domain Owner: Finance, Support & Admin (16 UCs, Security Interceptors)	ChatGPT (GPT-5.6), VS Code, Gemini 3.5 Pro
4	HE190169	Phạm Khương Duy	Domain Owner: Courses & Community (20 UCs, Frontend & Backend)	ChatGPT (GPT-4o), Cursor,Gemini Pro
5	HE181191	Bùi Đăng Cường	Domain Owner: AI, Chat & Marketplace (17 UCs, Realtime WebSocket)	ChatGPT (GPT-4o), Gemini, Antigravity

No.	SDLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed				
1	Requirement	Auth & Profile Domain Requirement Elicitation (17 Use Cases: Register, Verify OTP, Login JWT, Forgot/Reset Password, User/Tutor Profile, Upload Certificates)	Gemini 1.5 Pro, ChatGPT	Brainstormed 17 structured User Stories, acceptance criteria, and edge cases for authentication and tutor profile verification flows	Validated against business rules; adjusted password reset expiration rules and enforced 2-factor OTP verification for tutor registration	Group5_W1_Session1_AuthProfile_Stories.png / Google Drive Folder	17 User Stories, 34 Acceptance Criteria		AI suggested standard email verification without document upload state machine; student manually added PENDING_APPROVAL status transition				
2	Requirement	Search & Booking Domain Requirement Elicitation (18 Use Cases: Teaching Subjects, Tutor Search/Filter, Timeslot Management, Booking Lifecycle, Attendance)	Gemini 1.5 Pro, ChatGPT	18 detailed User Stories for tutor search filtering by subject/price, timeslot schedule grid, and booking state transitions	Refined booking lifecycle states (PENDING -> CONFIRMED -> COMPLETED/CANCELLED) and added auto-cancel cron job requirement for unpaid bookings	Group5_W1_Session2_Booking_Stories.png / Google Drive Folder	18 User Stories, 40 Acceptance Criteria		AI missed lesson attendance marking requirement; student added explicit LessonAttendance entity and confirmation flow				
3	Requirement	Finance, Support & Admin Domain Requirement Elicitation (16 Use Cases: Wallet Deposit/Withdraw, Booking Payment, Violation Reports, Support Tickets, Admin Dashboard)	Gemini 3.6 Pro, ChatGPT	16 User Stories covering e-wallet balance deduction, external payment gateway integration, report moderation, and admin system settings	Added strict idempotency requirement for wallet transactions to prevent double-charging during network retries	Group5_W1_Session3_FinanceAdmin_Stories.png / Google Drive Folder	16 User Stories, 30 Acceptance Criteria		AI generated generic payment refund logic; student customized refund policies based on booking cancellation timing rules				
4	Requirement	Courses & Community Domain Requirement Elicitation (20 Use Cases: Course Creation, Sessions/Materials, Open Classes, Assignments/Grading, Q&A Community Forum)	Gemini 1.5 Pro, ChatGPT	20 User Stories covering structured course curriculum, public vs tutor open classes, assignment submission grading, and Q&A upvote/downvote	Adjusted open class enrollment limits and added progress report calculation formula based on assignment scores	Group5_W1_Session4_CourseCommunity_Stories.png / Google Drive Folder	20 User Stories, 45 Acceptance Criteria		AI did not distinguish between Public Open Class and Tutor Class; student separated them into distinct business entities and flows				
5	Requirement	AI, Chat & Marketplace Domain Requirement Elicitation (17 Use Cases: Job Posts/Blubs, AI Assistant Chat, Bio Generator, Roadmap Generator, Realtime Websocket Chat)	Gemini 1.5 Pro, ChatGPT	17 User Stories for job bidding marketplace, AI tutor recommendation prompt requirements, roadmap generation, and Websocket chat rooms	Defined strict token usage limits (UsageLimitService) to prevent OpenAI API budget overrun and added STOMP/Websocket JWT auth specification	Group5_W1_Session5_AIChatMarketplace_Stories.png / Google Drive Folder	17 User Stories, 35 Acceptance Criteria		AI suggested unconstrained LLM calls; student designed quota tracking table (AIUsageLog) and daily rate limits				

No.	SDLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed			
1	System & Database Design	Conceptual & Logical ERD Design for Auth & Profiles (User, OtpVerification, PasswordResetOtp, TutorProfile, TutorVerification)	Antigravity, Gemini Pro	Suggested JPA entity schema and MySQL table attributes for users, OTP verification tokens, and tutor profile certifications	Normalised schema to 3NF; added unique constraints on email/phone, foreign key cascading rules, and indexed status fields	Group5_W2_Session1_Auth_ERD.png / draw.io link	5 Entities, 14 Columns indexed, 4 FK relationships		AI omitted timestamps and soft-delete flags; student added createdAt, updatedAt, and isDeleted columns across entities			
2	System & Database Design	Conceptual & Logical ERD Design for Search & Booking (Booking, BookingLog, BookingProposal, BookingSchedule, TimeSlot, Lesson, LessonAttendance, Subject, FavoriteTutor)	Antigravity, Gemini Pro	ERD schema connecting students and tutors through time slots, booking records, lesson schedules, and favorite lists	Designed composite keys for TimeSlot availability and added strict constraints to prevent double booking on the same TimeSlot ID	Group5_W2_Session2_Booking_ERD.png / draw.io link	8 Entities, 22 FK relationships, 3 indexes		AI missed booking historical audit trail; student created explicit BookingLog table to track every status modification with timestamps			
3	System & Database Design	Conceptual & Logical ERD Design for Finance, Support & Admin (Payment, Transaction, IdempotencyKey, SupportTicket, Report, SystemSetting, AdminActionLog)	Gemini 1.5 Pro, ChatGPT	Database schema for immutable financial transactions, e-wallet balances, idempotency keys, support tickets, and admin audit logs	Added optimistic locking version column (@Version) on wallet balance and unique index on (idempotencyKey key, userId)	Group5_W2_Session3_Finance_ERD.png / draw.io link	7 Entities, 18 Columns, Optimistic Locking structure		AI suggested mutable transaction records; student enforced append-only immutable ledger design for all financial transactions			
4	System & Database Design	Conceptual & Logical ERD Design for Courses & Community (Course, CourseSession, CourseEnrollment, TutorClass, OpenClass, Material, Question, Answer, Review, ProgressSupport)	Gemini 1.5 Pro	Schema for hierarchical course content (Course -> CourseSession -> Material), enrollments, Q&A threads, and rating reviews	Structured self-referencing hierarchy where needed and added unique constraint on CourseEnrollment(courseId, studentId)	Group5_W2_Session4_Course_ERD.png / draw.io link	10 Entities, 25 FK relationships, 4 indexes		AI created flat structure for questions/answers; student structured Question and Answer entities with accepted answer foreign key			
5	System & Database Design	Conceptual & Logical ERD Design for AI, Chat & Marketplace (jobPost, subApplication, AIChatSession, AIChatMessage, AIUsageLog, Notification, Conversation, Message)	Gemini 1.5 Pro	Database tables for job marketplace postings/bidding, real-time chat conversations/messages, AI sessions, and token usage tracking	Added composite indexes on Message(conversationId, createdAt) for fast chat history retrieval and AIUsageLog(userId, date) for daily quotas	Group5_W2_Session5_AIChat_ERD.png / draw.io link	8 Entities, 20 FK relationships, 5 composite indexes		AI did not include read receipt status on WebSocket messages; student added isRead and readAt timestamp fields to Message table			

No.	SDLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed				
1	UML Modeling	Use Case Diagram & Screen Flow Diagramming for Authentication & Profiles (Authentication_UCD.drawio, Profile_UCD.drawio, ScreenFlow_Authentication.drawio)	ChatGPT	UML Use Case diagrams and Screen Flow transitions connecting Login, Register, OTP verification modal, and Profile management screens	Standardized actor lifelines, added include/extend dependencies for OTP verification, and verified screen flow navigation paths against UI wireframes	deliverables/Authentication_UCD.drawio, Profile_UCD.drawio	2 Use Case Diagrams (17 UCs), 1 Screen Flow Diagram (8 screens)	5	AI generated colored boxes and ad-hoc shapes; student enforced strict monochrome black & white UML aesthetic and standardized syntax				
2	UML Modeling	Use Case Diagram & Screen Flow Diagramming for Search & Booking (Booking_UCD.drawio, Schedule_UCD.drawio, TutorDiscovery_UCD.drawio, ScreenFlow_Booking.drawio)	AntigravityChatGPT	UML Use Case diagrams for tutor search, timeslot scheduling, booking request creation, and lesson attendance screen transitions	Refined boundary boxes, clarified relationships between Student/Tutor actors, and aligned screen flow with actual frontend React routes	deliverables/Booking_UCD.drawio, Schedule_UCD.drawio	3 Use Case Diagrams (18 UCs), 2 Screen Flow Diagrams (12 screens)	5	AI omitted the automated cron job actor for unpaid bookings; student added System CronJob actor to the Use Case diagram				
3	UML Modeling	Use Case Diagram & Screen Flow Diagramming for Finance, Support & Admin (PaymentWallet_UCD.drawio, SupportReport_UCD.drawio, UC_Admin.drawio, ScreenFlow_PaymentandWallet.drawio)	Antigravity, Gemini Pro	UML Use Case and Screen Flow diagrams for e-wallet deposit/withdrawal, booking payment checkout, support ticket handling, and admin moderation	Separated Student/Tutor wallet views from Admin moderation views and verified screen transitions for external payment gateway callbacks	deliverables/PaymentWallet_UCD.drawio, UC_Admin.drawio	3 Use Case Diagrams (16 UCs), 2 Screen Flow Diagrams (10 screens)	4	AI generated overlapping transitions; student reorganized layout with clean right-angle connectors and exact screen labels				
4	UML Modeling	Use Case Diagram & Screen Flow Diagramming for Courses & Community (Learning_UCD.drawio, OpenClass_UCD.drawio, Community_UCD.drawio, ScreenFlow_CoursesandClasses.drawio)	Antigravity, ChatGPT	UML diagrams covering course enrollment, public/tutor open class management, assignment submission/grading, and Q&A community interactions	Ensured all 20 Use Cases were explicitly mapped across the 3 diagrams without cluttering and verified screen flows for assignment grading	deliverables/Learning_UCD.drawio, Community_UCD.drawio	3 Use Case Diagrams (20 UCs), 2 Screen Flow Diagrams (14 screens)	5	AI merged course creation and open class creation; student separated them into distinct use case modules for clear architectural separation				
5	UML Modeling	Use Case Diagram & Screen Flow Diagramming for AI Chat & Marketplace (AI_UCD.drawio, JobMarketplace_UCD.drawio, Chat_UCD.drawio, Notification_UCD.drawio, ScreenFlow_Chat.drawio)	Antigravity, ChatGPT	UML diagrams for job post creation/bidding, AI Assistant interaction, roadmap generation, real-time WebSocket chat, and notification popup flows	Added explicit boundary boxes for AI Orchestrator Service vs OpenAI REST API provider and detailed WebSocket chat room screen flows	deliverables/AI_UCD.drawio, Chat_UCD.drawio	4 Use Case Diagrams (17 UCs), 2 Screen Flow Diagrams (10 screens)	5	AI missed notification bell badge real-time state updates; student added explicit screen state transitions for unread/read notification states				

No.	SDLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed						
1	Detailed Class & Schema Design	Class Diagrams & JPA Entity Implementation for Auth & Profiles (User.java, OrgVerification.java, PasswordResetOp.java, TutorProfile.java, TutorVerification.java)	Antigravity, ChatGPT	Spring Boot JPA entity code with annotations (@Entity, @Table, @Column, @OneToMany) and Bean Validation rules (@NotBlank, @Email)	Reviewed and added custom DTO mappers, enum type converters, and repository interface signatures (UserRepository.java, OrgVerificationRepository.java)	backend/src/main/java/com/tutorfinder/backend/entity/User.java	5 JPA Entities, 5 Repositories, 380 LOC	5	Copilot generated eager loading (@ManyToOne(fetch = Eager)) by default; student changed all associations to FetchType LAZY to prevent N+1 performance bottlenecks						
2	Detailed Class & Schema Design	Class Diagrams & JPA Entity Implementation for Search & Booking (Booking.java, BookingLog.java, TimeSlot.java, Lesson.java, Subject.java, FavoriteTutor.java)	Antigravity, ChatGPT	JPA entities and repository interfaces with custom Spring Data JPA finder queries for booking availability and timeslot schedules	Implemented optimistic locking on Booking.java, added explicit foreign keys with indexing, and built custom query methods in BookingRepository.java	backend/src/main/java/com/tutorfinder/backend/entity/Booking.java	9 JPA Entities, 9 Repositories, 520 LOC	5	Copilot generated basic findById/studentId without join fetching; student wrote custom @Query with JOIN FETCH to optimise booking list queries						
3	Detailed Class & Schema Design	Class Diagrams & JPA Entity Implementation for Finance, Support & Admin (Payment.java, Transaction.java, Membership.java, SupportTicket.java, Report.java, SystemSetting.java)	Antigravity, ChatGPT	JPA entity classes for payments, transactions, support tickets, reports, and idempotency keys along with Spring Data repositories	Enforced @Column(updatable = false) on Transaction fields for immutability and implemented custom finder methods in MembershipRepository.java	backend/src/main/java/com/tutorfinder/backend/entity/Payment.java	7 JPA Entities, 7 Repositories, 410 LOC	5	Copilot missed unique constraint check on transaction reference IDs; student added explicit database unique constraints and JPA validation annotations						
4	Detailed Class & Schema Design	Class Diagrams & JPA Entity Implementation for Courses & Community (Course.java, CourseSession.java, CourseEnrollment.java, OpenClass.java, Material.java, Question.java, Answer.java, Review.java)	Antigravity, ChatGPT	JPA entities for courses, sessions, materials, open classes, Q&A community forum questions/answers, and tutor reviews with repository interfaces	Implemented cascading persistence rules where appropriate while preventing circular JSON serialization with @JsonIgnoreProperties and DTO projections	backend/src/main/java/com/tutorfinder/backend/entity/Course.java	10 JPA Entities, 10 Repositories, 640 LOC	5	Copilot generated bi-directional relationships causing infinite recursion during JSON serialization; student refactored to use clean DTO layers						
5	Detailed Class & Schema Design	Class Diagrams & JPA Entity Implementation for AI, Chat & Marketplace (JobPost.java, JobApplication.java, AIChatSession.java, AIChatMessage.java, AIUsageLog.java, Notification.java, Conversation.java, Message.java)	Antigravity, ChatGPT	JPA entity definitions for job marketplace postings/bids, AI chat sessions, usage limit tracking logs, and real-time conversation/message tables	Added index annotations on high-traffic queries (@Table(indexes = { @Index(name = "idx_msg_conv", columnList = "conversation_id") }))	backend/src/main/java/com/tutorfinder/backend/entity/JobPost.java	8 JPA Entities, 6 Repositories, 490 LOC	5	Copilot missed token usage quota constraint checking field types; student ensured AIUsageLog tokenized is tracked precisely as long with daily partitioning						

No.	SDLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed					
1	Dynamic Sequence Diagramming	UML Sequence Diagram & Controller/Service Scaffolding for Auth & Profile (SD_UC001_Login.drawio, SD_UC001_Register.drawio, SD_UC001_ResetPassword.drawio, AuthController.java, AuthService.java)	ChatGPT (GPT-4o)	Monochrome UML Sequence Diagrams mapping 1 to 1 with actual Spring Boot controller/service signatures and initial REST endpoint definitions	Enforced strictly black & white monochrome styling (#####0000000), prefixed lifelines with color (AuthController, AuthService), and aligned exact method verbs	deliverables/Sequence Diagram(s)/SD_UC001_Login.drawio	1 Monochrome Sequence Diagrams, 4 Controllers, 5 Services	5	AI generated fragmented activation boxes; student updated generator script to use continuous stack-tracked ranges spanning exactly from call to return					
2	Dynamic Sequence Diagramming	UML Sequence Diagram & Controller/Service Scaffolding for Search & Booking (SD_UC042_CreateBooking.drawio, SD_UC042_AcceptBooking.drawio, SD_UC042_CancelBooking.drawio, SD_UC030_CreateOpenClass.drawio, BookingController.java, BookingServiceImpl.java)	ChatGPT (GPT-4o)	UML Sequence Diagrams representing complete booking creation, tutor acceptance/cancellation, and open class workflows along with controller boilerplate	Verified exact method calls (validateBooking, checkAvailability, save) and ensured continuous stack-tracked activation rectangles with exact 2 layering	deliverables/Sequence Diagram(s)/SD_UC042_CreateBooking.drawio	4 Monochrome Sequence Diagrams, 6 Controllers, 6 Services	5	AI placed message arrows behind activation white fills; student enforced strict 2-order rendering so all arrowheads land snugly on top of activation bars					
3	Dynamic Sequence Diagramming	UML Sequence Diagram & Controller/Service Scaffolding for Finance, Support & Admin (SD_UC043_WalletPayment.drawio, SD_UC043_DemoPayment.drawio, SD_UC050_DepositWallet.drawio, SD_UC011_ApproveTutorApplication.drawio, WalletController.java, PaymentServiceImpl.java)	Gemini, ChatGPT	Sequence diagrams for wallet payment processing, demo checkout, deposit flow, and admin application approval along with controller/service scaffolding	Added explicit opt/alt UML fragments for payment gateway validation and verified exact DTO return messages on dashed return arrows	deliverables/Sequence Diagram(s)/SD_UC043_WalletPayment.drawio	4 Monochrome Sequence Diagrams, 11 Controllers, 6 Services	5	AI generated unescaped backticks in template strings for alt/loop/opt; student fixed template string escaping and verified exact diagram XMML output					
4	Dynamic Sequence Diagramming	UML Sequence Diagram & Controller/Service Scaffolding for Courses & Community (SD_UC060_SubmitAssignment.drawio, SD_UC060_GradeAssignment.drawio, SD_UC106_CreateQuestion.drawio, SD_UC106_AnswerQuestion.drawio, CourseController.java, CommunityServiceImpl.java)	ChatGPT (GPT-4o)	Sequence diagrams detailing student assignment submission, tutor grading, community Q&A question posting, and answering flow along with service interfaces	Ensured generous headroom (40px) inside loop/alt fragment headers so frame labels never overlap with inner message steps	deliverables/Sequence Diagram(s)/SD_UC060_SubmitAssignment.drawio	4 Monochrome Sequence Diagrams, 8 Controllers, 8 Services	5	AI generated colliding text labels inside fragment frames; student customized diagram generator layout engine to automatically allocate clean header margins					
5	Dynamic Sequence Diagramming	UML Sequence Diagram & Controller/Service Scaffolding for AI, Chat & Marketplace (SD_UC103_AIChat.drawio, SD_UC141_Recommendation.drawio, SD_UC142_RoadmapGenerator.drawio, SD_UC010_SubmitTutorApplication.drawio, AIChatController.java, AIOrchestratorService.java)	ChatGPT	Sequence diagrams for AI Assistant chat orchestration, AI tutor recommendation, roadmap generation, and tutor registration along with WebSocket config	Mapped exact orchestration calls between AIChatController -> AIOrchestratorService -> UsageLimitService -> OpenAIProviderService	deliverables/Sequence Diagram(s)/SD_UC103_AIChat.drawio	4 Monochrome Sequence Diagrams, 7 Controllers, 6 Services	5	AI missed token usage quota check step before calling external LLM; student added explicit validation call to UsageLimitService inside the sequence diagram					

No.	SLOC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed						
1	Core Backend Implementation	Business Logic Implementation for Auth & Profiles (AuthService.java, UserProfileService.java, CptgService.java, JwtUtils.java)	Antigrahy, ChatGPT	Implemented full JWT access/refresh token generation and validation, OTP generation/email verification logic, password hashing with BCrypt, and profile updating	Added strict input validation, custom exception handling (ResourceNotFoundException, UnauthorizedException), and transactional boundaries (@Transactional)	backend/src/main/java/com/tutorfinder/backend/service/impl/AuthServiceImpl.java	5 Core Services, 12 REST API Endpoints, 650 LOC	5	Copilot suggested insecure JWT secret key generation and hardcoded expiration; student externalized configuration to application.yml with strong HMAC verification						
2	Core Backend Implementation	Business Logic Implementation for Search & Booking (BookingServiceImpl.java, ScheduleService.java, TutorService.java, TutorAvailabilityService.java, SubjectServiceImpl.java)	Antigrahy, ChatGPT	Implemented complex tutor availability checking, booking creation with conflict detection, timeslot generation, schedule querying, and scheduled cron jobs	Added @Scheduled(cron = "0 * * * *") in BookingServiceImpl.java to automatically cancel unpaid pending bookings after 15 minutes and release timeslots	backend/src/main/java/com/tutorfinder/backend/service/impl/BookingServiceImpl.java	6 Core Services, 15 REST API Endpoints, 880 LOC	5	Copilot missed concurrency race condition during timeslot booking; student wrapped booking creation inside @Transactional with pessimistic/optimistic check						
3	Core Backend Implementation	Business Logic Implementation & Security Interceptors for Finance, Support & Admin (WalletServiceImpl.java, PaymentServiceImpl.java, AdminServiceImpl.java, IdempotencyInterceptor.java, ContentCachingFilter.java)	Gemini 1.5 Pro	Implemented e-wallet deposit/withdrawal/payment processing, admin moderation approval logic, custom idempotency interceptor, and request body caching filter	Enforced IdempotencyInterceptor to impact X-Idempotency-Key headers and check Redis/MemDB before executing payment deduction to guarantee exactly-once processing	backend/src/main/java/com/tutorfinder/backend/service/impl/WalletServiceImpl.java	6 Core Services, 2 Interceptors/Filters, 18 REST APIs, 790 LOC	5	Copilot generated filter that consumed HttpServletResponse input stream once causing subsequent controllers to fail; student implemented ContentCachingRequestWrapper						
4	Core Backend Implementation	Business Logic Implementation for Courses & Community (CourseServiceImpl.java, PublicOpenClassServiceImpl.java, MaterialServiceImpl.java, CommunityServiceImpl.java, ReviewServiceImpl.java)	Antigrahy	Implemented course catalog management, public/tutor open class enrollment logic, course material file storage handling, assignment grading, and ORA voting algorithms	Wrote custom rating calculation queries in ReviewServiceImpl.java to dynamically update tutor average rating and verified assignment submission status checks	backend/src/main/java/com/tutorfinder/backend/service/impl/CourseServiceImpl.java	8 Core Services, 20 REST API Endpoints, 920 LOC	5	Copilot generated unoptimized loop for recalculating tutor ratings; student wrote single aggregated SQL query inside ReviewRepository to update tutor stats instantly						
5	Core Backend Implementation	Business Logic Implementation for AI, Chat & Marketplace (AIOrchestratorService.java, UsageLimitService.java, OpenAIProviderService.java, ChatServiceImpl.java, WebSocketConfig.java, WebSocketChatInterceptor.java)	ChatGPT	Implemented AI orchestrator for structured chat/RLS/roadmap generation, OpenAI REST integration, daily token quota tracking, and STOMP WebSocket chat messaging	Built custom WebSocketGatewayInterceptor.java to intercept STOMP CONNECT headers, validate JWT tokens before opening WebSocket sessions, and route messages securely	backend/src/main/java/com/tutorfinder/backend/service/impl/AIOrchestratorServiceImpl.java	6 Core Services, 2 WebSocket Configs/Interceptors, 14 REST APIs, 850 LOC	5	AI suggested blocking synchronous REST calls to OpenAI; student added timeout handling, circuit breaker logic, and fallbacks when LLM API quota is exceeded						

No.	SDLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed			
1	Frontend UI Integration	Frontend React UI Implementation for Auth & Profiles LoginPage.jsx, RegisterPage.jsx, ProfilePage.jsx, OtpModal.jsx	v0 by Vercel, Gemini Pro	Responsive React components with Tailwind/Vanilla CSS styling, form validation using React Hook Form, and Axios JWT interceptor integration	Integrated refresh token silent rotation in Axios interceptor and added loading state micro-animations for better UX	frontend/src/pages/auth/LoginPage.jsx	4 Pages, 6 Components, 580 LOC	5	v0 generated static mock state; student wired complete API service layer with error handling and toast notifications			
2	Frontend UI Integration	Frontend React UI Implementation for Search & Booking TutorSearchPage.jsx, BookingCalendar.jsx, BookingCheckoutModal.jsx, MyBookingsPage.jsx	v0 by Vercel, Gemini Pro	Interactive tutor search filters (subject, price, rating), interactive time slot grid selection calendar, and booking lifecycle management cards	Customized calendar grid rendering to handle timezone conversions accurately and added immediate state reflection on booking cancellation	frontend/src/pages/booking/BookingCalendar.jsx	5 Pages, 8 Components, 740 LOC	5	v0 generated calendar grid without responsive mobile layout; student refactored grid to flex-wrap cards on smaller viewports			
3	Frontend UI Integration	Frontend React UI Implementation for Finance, Support & Admin WalletPage.jsx, DepositModal.jsx, SupportTicketPage.jsx, AdminDashboardPage.jsx, AdminVerificationTable.jsx	v0 by Vercel, Gemini Pro	E-wallet balance card, deposit/withdraw modals, transaction history table with pagination, support ticket dialogs, and admin moderation dashboard	Implemented real-time balance update check after external payment gateway return and built multi-action approval/rejection buttons for admin	frontend/src/pages/wallet/WalletPage.jsx	6 Pages, 10 Components, 680 LOC	5	v0 created static tables without sorting/filtering; student integrated dynamic pagination, status filtering, and search bar			
4	Frontend UI Integration	Frontend React UI Implementation for Courses & Community CourseListPage.jsx, CourseDetailPage.jsx, OpenClassPage.jsx, CommunityPage.jsx, AssignmentModal.jsx	v0 by Vercel, Gemini Pro	Course catalog listing, detailed curriculum viewer, open class registration cards, assignment file upload drozone, and community Q&A forum threads	Implemented markdown rendering for Q&A answers with code highlighting and built drag-and-drop file upload progress bar for materials	frontend/src/pages/courses/CourseListPage.jsx	6 Pages, 9 Components, 790 LOC	5	v0 generated basic textarea for Q&A; student integrated rich markdown editor and upvote/downvote optimistic UI updates			
5	Frontend UI Integration	Frontend React UI Implementation for AI, Chat & Marketplace JobMarketplacePage.jsx, JobBidModal.jsx, AIChatWidget.jsx, RoadmapPage.jsx, ChatBox.jsx	v0 by Vercel, Gemini Pro	job marketplace posting/bidding board, floating AI Assistant chat widget with streaming text effect, interactive learning roadmap visualizer, and WebSocket chat room	Connected STOMP WebSocket client inside ChatBox.jsx to receive instant messages and render unread notification badges dynamically without page refresh	frontend/src/pages/chat/ChatBox.jsx	5 Pages, 8 Components, 810 LOC	5	v0 did not handle WebSocket reconnection logic; student added exponential backoff/reconnection hook when network drops			

No.	SOLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed				
1	Performance & Security Audit	Security & Performance Audit for Auth & Profiles (BCrypt tuning, JWT signature check, profile storage sanitization)	ChatGPT (GPT-4o)	Security analysis and recommendations for BCrypt work factor tuning (strength 12), JWT algorithm enforcement (HS512), and file upload path sanitization	Implemented strict path traversal prevention on profile/certificate uploads and verified CORS configuration across frontend/backend endpoints	backend/src/main/java/com/tutorfinder/backend/config/SecurityConfig.java	3 Security Policies enforced, 100% path sanitized	5	AI suggested permissive CORS wildcard (*) during development; student locked down CORS allowed origins strictly to production/training frontend URLs				
2	Performance & Security Audit	Database Query Optimization & indexing for Search & Booking (BookingRepository.java, TimeSlotRepository.java)	Antigravity	Optimized JPA queries using @Query with JOIN FETCH to eliminate N+1 query overhead when loading booking lists and tutor schedules	Added composite indexes on Booking(tutor_id, status) and TimeSlot(tutor_id, start_time) to speed up availability checking under load	backend/src/main/java/com/tutorfinder/backend/repository/BookingRepository.java	4 Custom Queries optimized, 3 indexes created	5	Copilot generated subqueries that resulted in full table scans; student refactored to indexed join queries reducing response time by 75%				
3	Performance & Security Audit	Security Auditing & Audit Trail Verification for Finance, Support & Admin (ContentCachingFilter, AdminActionLoggingRepository, CSRF protection)	Gemini, ChatGPT	Audited financial request validation, ensured sensitive balance adjustments are logged immutably in AdminActionLog, and verified rate limiting on payment APIs	Implemented custom request body inspection filter ensuring all financial mutations carry valid signatures and exact timestamps	backend/src/main/java/com/tutorfinder/backend/filter/ContentCachingFilter.java	2 Security Filters audited, 100% financial mutations logged	5	AI missed checking integer overflow on e-wallet balance deposits; student enforced BigDecimal/strict range checking on all financial inputs				
4	Performance & Security Audit	Query Pagination & Cache Tuning for Courses & Community (CourseRepository.java, QuestionRepository.java, OpenCourseRepository.java)	Antigravity	Implemented Spring Data Pageable across all public course listing, open class browsing, and community Q&A search endpoints to prevent OOM errors	Added HTTP caching headers and ETag support on static course curriculum materials to reduce redundant server bandwidth consumption	backend/src/main/java/com/tutorfinder/backend/controller/CourseController.java	8 Paginated Endpoints, ETag Caching enabled	5	Copilot generated unpaginated findAll() for open class enrollments; student enforced strict page size limits (max 50 per page)				
5	Performance & Security Audit	AI Token Optimization & WebSocket Rate Limiting for AI Chat & Marketplace (OpenAIProviderService.java, WebSocketConfig.java)	ChatGPT (GPT-4o)	Optimized system prompt lengths for AI bot and roadmap generation to reduce OpenAI prompt token cost by 40% and implemented WebSocket message rate throttling	Added sliding window rate limiter on STOMP message broadcasting to prevent chat room spamming and DOS attacks on real-time connections	backend/src/main/java/com/tutorfinder/backend/service/impl/OpenAIProviderServiceImpl.java	40% Token savings achieved, WebSocket throttle active	5	AI suggested verbose system prompts exceeding 2000 tokens; student engineered compact, high-precision JSON instruction schemas				

No.	SDLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed						
1	Unit & Integration Testing	Unit & Integration Testing for Auth & Profiles (AuthServiceTest.java, UserServiceTest.java, AuthUtilsTest.java)	Antigravity, ChatGPT	Unit 5 & Mockito test cases verifying user registration, OTP validation logic, JWT expiration handling, and unauthorized access rejections	Student's Validation / Modification Achieved 94% instruction coverage on authentication domain; verified edge cases such as expired OTPs and duplicate email registration attempts	backend/src/test/java/com/tutorfinder/backend/service/AuthServiceTest.java	18 June 5 Test Cases, 94% Coverage	5	Copilot generated tests without verifying mock invocation; student added verify(userRepository, times(1)).save(any()) assertions across all suites						
2	Unit & Integration Testing	Unit & Integration Testing for Search & Booking (BookingServiceImpTest.java, SchedulerServiceTest.java, TravelerServiceTest.java)	Antigravity	Comprehensive test suites checking booking availability logic, double booking prevention, schedule overlapping validation, and cron job execution	Simulated concurrent booking requests using ExecutorService to verify transaction isolation and verified auto-cancel logic with mocked clocks	backend/src/test/java/com/tutorfinder/backend/service/BookingServiceImpTest.java	22 June 5 Test Cases, 91% Coverage	5	Copilot missed testing concurrent race conditions; student wrote multi-threaded integration test to confirm booking behavior						
3	Unit & Integration Testing	Unit & Integration Testing for Finance, Support & Admin (WalletServiceImpTest.java, PaymentServiceImpTest.java, IdempotencyInterceptorTest.java)	Gemini Pro	Test cases verifying wallet deposit/withdrawal calculation accuracy, insufficient balance rejection, idempotency key replay prevention, and admin action logging	Verified that duplicate requests with the same X-Idempotency-Key return cached responses without executing wallet deduction twice	backend/src/test/java/com/tutorfinder/backend/service/WalletServiceImpTest.java	16 June 5 Test Cases, 93% Coverage	5	AI missed testing rollback behavior on transaction failure; student added explicit test verifying wallet balance rollback when external gateway fails						
4	Unit & Integration Testing	Unit & Integration Testing for Courses & Community (CourseServiceImpTest.java, CommunityServiceImpTest.java, RecommendationTest.java)	Antigravity	Test suites validating course enrollment permissions, assignment grading calculation, Q&A accepted answer toggle logic, and tutor rating updates	Verified that marking an answer as accepted correctly updates question status and that review score averages compute accurately across edge cases	backend/src/test/java/com/tutorfinder/backend/service/CoursesServiceImpTest.java	20 June 5 Test Cases, 90% Coverage	5	Copilot missed edge case of zero reviews when dividing score total; student verified division by zero safety check inside review calculation						
5	Unit & Integration Testing	Unit & Integration Testing for AI Chat & Marketplace (AIOrchestratorServiceTest.java, UsageLimitServiceTest.java, ChatServiceImpTest.java)	ChatGPT	Test suites mocking OpenAI REST API responses, verifying daily token quota limits, chat session persistence, and WebSocket message dispatching	Simulated rate limit exceeded exceptions from OpenAI provider and verified clean fallback response generation without crashing the backend service	backend/src/test/java/com/tutorfinder/backend/service/AIOrchestratorServiceTest.java	16 June 5 Test Cases, 89% Coverage	5	Copilot struggled to mock complex WebClient REST responses; student wrote clean WebClient.MockWebServer helper utilities to test LLM integrations reliably						

No.	SDLC Phase	Task / Activity	AI Tool Used	AI Output	Student's Validation / Modification	Evidence / Link	Quantitative Measure	Value Added (1-5)	Risks / Limitations Observed					
1	Final Verification & Traceability	Final Traceability & Documentation Verification for Auth & Profiles (Final_TutorFinder_SD5_FORMATTED.docx, FINAL_USER_MANUAL_FORMATTED.docx, Template1/4/5)	ChatGPT (GPT-4o)	Verified 100% traceability between User/Auth/Profile source code, 17 Use Cases, class specifications, and WF-01/WF-02/WF-06 user manual sections.	Cross-checked all method signatures and database table structures against final release documents, ensuring zero discrepancies across the package.	deliverables/final doc/Final_TutorFinder_SD5_FORMATTED.docx	17 Use Cases fully traced, 100% verified	5	AI summarized manual sections without verifying exact UI button labels; student proofread all workflow steps against actual running React app screens.					
2	Final Verification & Traceability	Final Traceability & Monochrome Sequence Diagram Verification for Search & Booking (20 Monochrome Sequence Diagrams, SequenceDiagram_Specification.md, Overview.html)	ChatGPT	Verified all 20 monochrome Draw.io Sequence Diagrams against exact backend Controller/Service/Repository signatures with : prefixed lifelines and exact Z-order.	Secured generate_sequence_diagrams.js to confirm clean two-pass simulation, continuous stack-tracked activation bars, and zero text collisions.	deliverables/Sequence Diagrams() /SequenceDiagram_Specification.md	20 Monochrome Sequence Diagrams, 100% code-verified	5	AI generated static diagrams; student maintained automated generator script as single source of truth guaranteeing perfect diagram-to-code alignment.					
3	Final Verification & Traceability	Final Traceability & RDS/Issues Verification for Finance, Support & Admin (Final_TutorFinder_RDS_FORMATTED.docx, Template4_IssuesReport.xlsx, Useremptokeny specs)	Gemini 1.5 Pro, ChatGPT	Verified database table descriptions, security interceptor documentation, and closed all defect reports in Template4_IssuesReport.xlsx	Confirmed that all 16 Use Cases for wallet, support, and admin dashboard match exact Controller mappings in 05_RDS_Code_Mapping.xlsx	deliverables/final doc/Final_TutorFinder_RDS_FORMATTED.docx	16 Use Cases verified, 0 Open Defect Issues	5	AI generated generic table descriptions; student verified exact data types, constraints, and optimistic locking annotations directly from JPA entities.					
4	Final Verification & Traceability	Final Traceability & User Manual Verification for Courses & Community (WF-05 to WF-12 in FINAL_USER_MANUAL_FORMATTED.docx, Course/Community SD5 sections)	ChatGPT (GPT-4o)	Verified step-by-step user manual workflows for course enrolment, open class scheduling, assignment grading, and Q&A community interactions.	Ensured all 20 Use Cases and 30 JPA entities for courses and community are documented accurately across Template 2 (RDS) and Template 3 (SDS).	deliverables/final doc/FINAL_USER_MANUAL_FORMATTED.docx	20 Use Cases verified, 8 User Manual Workflows proofread	5	AI missed documenting assignment file size limit in User Manual; student added explicit instruction note regarding 25MB attachment limit.					
5	Final Verification & Traceability	Final Traceability & AI Usage Report Compilation for AI, Chat & Marketplace (Template0_SWP391_AI_Usage_Report_Template.xlsx, WF-14 to WF-22 in User Manual)	ChatGPT (GPT-4o)	Compiled and verified the complete load-balanced AI Usage Report Template 0 mapping of 5 team members to their exact Business Domains and 88 total Use Cases.	Verified that every single AI activity log entry reflects quantitative metrics, student engineering validation, and clear value-add ratings.	deliverables/final doc/Template0_SWP391_AI_Usage_Report_Template.xlsx	88 Total Use Cases accounted for, 5 Domain Owners verified	5	AI generated generic summary rows; student structured comprehensive load-balanced documentation exactly matching TEAM WORKLOAD PLANNING.docx					

Column	Description & How to Fill							
No.	Sequential number of the log entry (1, 2, 3...).							
SDLC Phase	Which phase of the project this AI usage belongs to (e.g., Requirement, Design, Implementation, Testing, Reporting).							
Task / Activity	Specific task you used AI for (e.g., User Story Development, ERD Design, Workflow 1 Coding, Test Case Generation).							
AI Tool Used	Name of the AI tool (e.g., ChatGPT, GitHub Copilot, Copilot X, Bard, CodeWhisperer).							
AI Output	The actual output AI produced (summary only, e.g., "10 user stories generated", "ERD with 6 entities", "sample Spring Boot controller").							
Student's Validation	How you validated or modified the AI output (e.g., "kept 5 user stories, rewrote 3", "adjusted foreign keys", "fixed syntax errors", "refined test cases").							
Evidence / Link	Link to a shared Google Drive folder containing screenshots/videos of the AI chat session(s). Naming convention: GroupX_SessionY_Activity (e.g., Group5_Session2_UserStory.png).							
	Each screenshot/videos must clearly show: prompt, AI response, and student follow-up.							
Quantitative Metrics	Numerical evidence of the contribution (e.g., number of user stories, number of entities, number of classes/methods, LOC, number of test cases).							
Value Added (1-5)	Self-assessment of AI's usefulness (1 = not useful, 5 = very useful).							
Risks / Limitations	What limitations or risks you observed when using AI (e.g., "generated irrelevant user stories", "missing validation logic", "incomplete test coverage").							